

Question	Answer	Marks
10(a)(i)	penetration of beam	M1
	greater hardness means greater penetration/shorter wavelength/higher frequency/higher photon energy	A1
10(a)(ii)	greater accelerating potential difference or greater p.d. between anode and cathode	B1
10(b)	$I = I_0 \exp(-\mu x)$ ratio = $(\exp \{-1.5 \times 2.9\}) / (\exp \{-4.0 \times 0.95\}) (= \exp \{-0.55\})$	C1
	= 0.58	A1

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11(c)	electrons (in gas atoms/molecules) interact with photons	B1